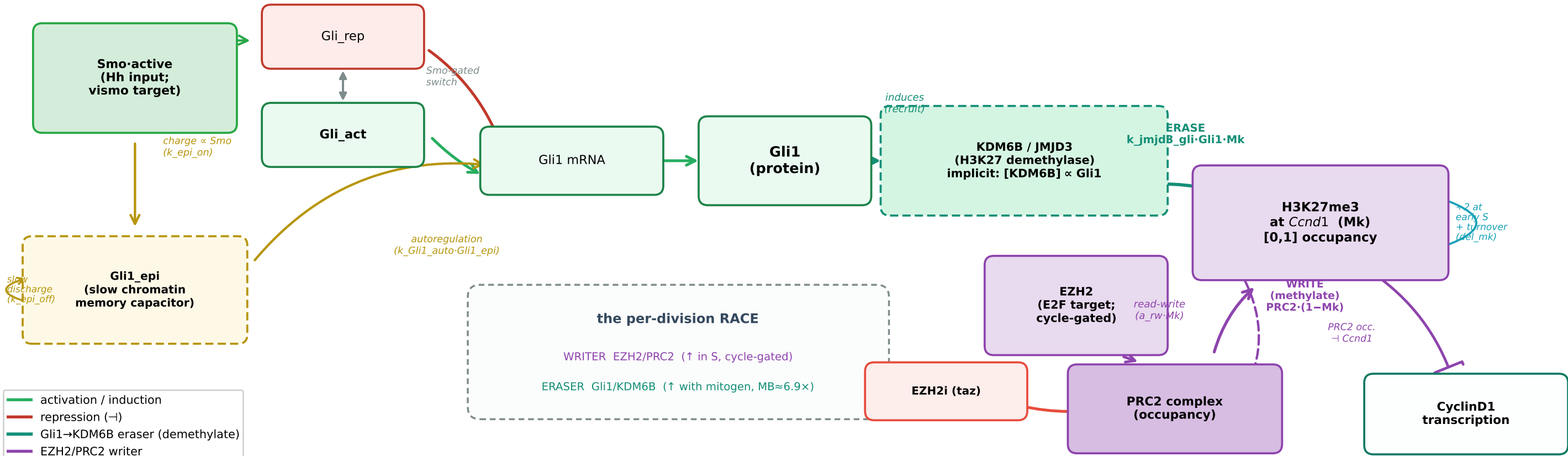


Gli1 → Kdm6b/Jmjd3 eraser of H3K27me3 at *Ccnd1* — detailed wiring + equations (v44.1)

mitogen-gated ERASER (Gli1 → KDM6B/JMJD3 demethylase) vs cycle-gated WRITER (EZH2/PRC2) — the per-division race that sets the mark



① Gli1 axis (mitogen input, with slow autoregulatory memory)

$$\begin{aligned} \text{Gli_rep} &\rightleftharpoons \text{Gli_act} \quad (\text{Smo-gated activator/repressor switch}) \\ d[\text{Gli1_mRNA}]/dt &= V_{\text{max}} \cdot \text{Gli_act}^2 / (K_{\text{GA}}^2 + \text{Gli_act}^2) \cdot (1 - \text{Gli_rep}^2 / (K_{\text{GR}}^2 + \text{Gli_rep}^2)) \cdot \gamma_{\text{Ptch}} \\ &\quad + k_{\text{Gli1_auto}} \cdot \text{Gli1_epi} - k_{\text{mdeg}} \cdot \text{Gli1_mRNA} \\ d[\text{Gli1_epi}]/dt &= k_{\text{epi_on}} \cdot (1 - \text{Ptch}) \cdot \text{Smo}^4 / (K_{\text{auto}}^4 + \text{Smo}^4) \cdot (1 - \text{Gli1_epi}) - k_{\text{epi_off}} \cdot \text{Gli1_epi} \\ d[\text{Gli1}]/dt &= k_{\text{Gli1_tl}} \cdot \text{Gli1_mRNA} - k_{\text{Gli1_deg}} \cdot \text{Gli1} \end{aligned}$$

② Eraser (KDM6B/JMJD3, implicit)

KDM6B is NOT a state variable.
Quasi-static: $[\text{KDM6B}] \propto \text{Gli1}$,
folded into $k_{\text{jmd3_gli}}$.

$$\begin{aligned} \text{erase flux} &= \\ k_{\text{jmd3_gli}} \cdot \text{Gli1} \cdot \text{Mk} \end{aligned}$$

③ H3K27me3 (Mk) balance at the *Ccnd1* domain

$$\begin{aligned} d[\text{Mk}]/dt &= \underbrace{\text{PRC2} \cdot (1 - \text{Mk})}_{\text{EZH2/PRC2, cycle}} - \underbrace{\text{del_mk} \cdot \text{Mk}}_{\text{Gli1/KDM6B, mitogen}} - \underbrace{k_{\text{jmd3_gli}} \cdot \text{Gli1} \cdot \text{Mk}}_{\text{ERASE}} \\ \text{discrete: at Dna} &> 0.05 \rightarrow \text{Mk} \leftarrow \frac{1}{2} \cdot \text{Mk} \quad (\text{replicative dilution, once per S}) \\ \text{PRC2} &= \text{EZH2} \cdot (1 - \text{EZH2i}) \cdot (a_0 + a_{\text{rw}} \cdot \text{Mk}) \cdot (1 - g \cdot \text{Cd_mRNA}^p / (K_{\text{tx}}^p + \text{Cd_mRNA}^p)) \end{aligned}$$

④ Baked parameters (v44.1) + net effect

$$\begin{aligned} k_{\text{jmd3_gli}} &0.0478 \quad \text{del_mk} \quad 0.0015 \\ a_0 &3.78\text{e-}4 \quad a_{\text{rw}} \quad 4.30\text{e-}3 \quad (a_{\text{rw}}/a_0 \quad 11.4) \\ g_{\text{prc2}} &0.0424 \quad K_{\text{prc2}} \quad 3.5\text{e-}3 \quad n \quad 3.39 \\ f_0_{\text{prc2}} &0.0505 \quad K_{\text{tx}} \quad 5.12 \quad p_{\text{tx}} \quad 3.27 \\ \text{Gli: } V_{\text{max}} &1.27 \quad \text{auto} \quad 0.012 \quad \text{on/off} \quad 0.02/8\text{e-}4 \end{aligned}$$

MB Gli1 ≈ 6.9× GNP → erase ≈ 6.9× stronger → ChIP *Ccnd1* mark MB/GNP ≈ 0.55 (MB < GNP)

Net over divisions: *Mk* accumulates when $\text{WRITE} > (\text{turnover} + \text{ERASE} + \frac{1}{2}\text{-dilution})$ each cycle; MB's high *Gli1* tilts the race toward erasure despite higher *EZH2*.